

Haonan Wu

Ph.D. Student/Research Assistant
Computer Science and Engineering at Penn State
State College, PA, USA

CONTACT INFORMATION

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EDUCATION

The Pennsylvania State University (PSU), USA Aug. 2023 - present
Ph.D. in Computer Science and Engineering **Research Topics:** Theoretical and Algorithmic Bioinformatics
Supervisor: Prof. Paul Medvedev

Shandong University (SDU), PRC Sep. 2020 - Jun. 2023
Master of Science in Operational Research and Cybernetics
GPA: 92.17/100
Relevant Coursework: Graphs and Digraphs, Combinatorial Optimization, Analysis and Design of Algorithms, Modern Functional Analysis, Elements of Modern Algebra, Mathematical Analysis

Shandong University (SDU), PRC Sep. 2016 - Jun. 2020
Bachelor of Science in Biotechnology
GPA: 90.65/100
Relevant Coursework: Biochemistry, Cell Biology, Molecular Biology

WORK EXPERIENCE

The Pennsylvania State University State College, USA
Research Assistant in Department of Computer Science and Engineering Jan. 2024 – present
Supervisor: **Supervisor:** Prof. Paul Medvedev

Developing bioinformatics theories and algorithms.

RESEARCH INTERESTS

I am interested in developing theories, algorithms, and practical tools for bioinformatics and addressing biological problems through these methods. My research focuses on

1. Design methods with solid math/algorithms or at least mathematical intuition.
2. Build practically useful tools with heuristics benefiting the bio/bioinfo communities.

Currently, I am working on theories and estimators of mutation and evolutionary processes based on k -mers, particularly on repetitive genomic sequences.

RESEARCH EXPERIENCE

Repeat-robust estimators of substitution rates 2026

We present three new estimators that are robust to the presence of repeats. The estimators are applicable in different settings, based on whether they need count information from zero, one, or both of the sequences. We evaluate our estimators empirically using highly repetitive alpha satellite sequences. Our estimators each perform best in their class, and our strongest estimator outperforms all other tested estimators.

Software:

- <https://github.com/medvedevgroup/Accurate-repeat-aware-kmer-based-estimator>

A k-mer-based estimator of the substitution rate between repetitive sequences 2025

We relax the non-repetitive assumption and propose a novel estimator for the mutation rate. We derive theoretical bounds on our estimator's bias. Our experiments show that it remains accurate for repetitive genomic sequences, such as the alpha satellite higher order repeats in centromeres. We demonstrate our estimator's robustness across diverse datasets like the HOR region of the centromere.

Softwares:

- <https://github.com/medvedevgroup/Repeat-Aware-Substitution-Rate-Estimator>
- <https://github.com/Wu-Haonan/Repeat-Aware-mutation-rate-estimator>
- <https://github.com/Wu-Haonan/Repeat-Aware-phylogenetic-distance-estimator>

Spatom: a graph neural network for structure-based protein-protein interaction site prediction 2023

We propose Spatom, a novel framework for PPI site prediction. This framework first defines a weighted digraph for a protein structure to precisely characterize the spatial contacts of residues, then performs a weighted digraph convolution to aggregate both spatial local and global information and finally adds an improved graph attention layer to drive the predicted sites to form more continuous region(s). We tested it on a diverse set of challenging protein-protein complexes and demonstrated the best performance among all the compared methods.

Softwares:

- Web server: <http://liulab.top/Spatom/server>
- Source code: <https://github.com/Wu-Haonan/Spatom>

PEER REVIEWED PUBLICATIONS

Under review

- **Haonan Wu**, Antonio Blanca, and Paul Medvedev, A k-mer-based estimator of the substitution rate between repetitive sequences, 2026 (submit to Algorithms for Molecular Biology)
- **Haonan Wu**, and Paul Medvedev, The gift of creation: repeat-robust estimators of substitution rates, 2026 (submit to Intelligent Systems for Molecular Biology)

Preprints

- **Haonan Wu**, Antonio Blanca, and Paul Medvedev, A k-mer-based estimator of the substitution rate between repetitive sequences, *bioRxiv* 2025.06.19.660607; doi: <https://doi.org/10.1101/2025.06.19.660607>

Journals

- **Haonan Wu**, Jiyun Han, Shizhuo Zhang, Gaojia Xin, Chaozhou Mou, Juntao Liu, Spatom: a graph neural network for structure-based protein-protein interaction site prediction, *Briefings in Bioinformatics*, Volume 24, Issue 6, November 2023, bbad345, <https://doi.org/10.1093/bib/bbad345>

Conference proceedings

- **Haonan Wu**, Antonio Blanca, and Paul Medvedev. A k-mer-Based Estimator of the Substitution Rate Between Repetitive Sequences. In 25th International Conference on Algorithms for Bioinformatics (WABI 2025). Leibniz International Proceedings in Informatics (LIPIcs), Volume 344, pp. 20:1-20:20, Schloss Dagstuhl – Leibniz-Zentrum für Informatik (2025) <https://doi.org/10.4230/LIPIcs.WABI.2025.20>

PROFESSIONAL SERVICE

Peer Reviewer

Journals: Heliyon

Conferences: RECOMB (2025, 2026), ISMB (2025, 2026), ACM-BCB (2025)

Short courses

- Deep-learning workshop, School of Mathematics and Statistics, SDU, China Winter 2022
Handouts of the course.

Undergraduate Mentor

- Inorganic and Analytical Chemistry, School of Marine, SDU, China Fall 2018
- Probability and Mathematical Statistics II, School of Marine, SDU, China Spring 2019

TALKS & POSTER

Talks

- Title: Don't Repeat 'No Repeats':
A k -mer-based estimator of the substitution rate between repetitive sequences
Genome Informatics 2025, Cold Spring Harbor Laboratory, NY, USA Nov. 5-8 2025
- Title: Don't Repeat 'No Repeats':
A k -mer-based estimator of the substitution rate between repetitive sequences
WABI 2025, University of Maryland, MD, USA Aug. 20 2025
- Topic: Applications of Digraph in bioinformatics and computational biology
Graphs and Digraphs, School of Mathematics, SDU, China Spring 2022
- Topic: Introduction of Single Molecular Sequence and assembly algorithms for long-reads
Sequence and Assembly Seminar, School of Mathematics and Statistics, SDU, China Fall 2019

Posters

- WABI 2025, University of Maryland, MD, USA (Link)
- Rao Prize Conference 2025, Penn State University, PA, USA (Link)

OTHER EXPERIENCES

Explored the possible reason for abundance of oxides on the Martian surface Jun. 2017 - Oct. 2018
Planetary Lab, Space Science School, SDU with Dr. Zhongchen Wu

SKILLS

- **Programming:** Python, MATLAB, C++, experience programming in Linux server
- **Package:** Pytorch, scikit-learn
- **Scripting Language:** $\text{\LaTeX}$
- **Language:** Chinese (native speaker), English (working language)

REFERENCES

Prof. Paul Medvedev

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Prof. Antonio Blanca

Associate Professor

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